Universitas Pandrosion: Part VIII
The Geometric-Decreasing Start Strategy
Empirical Phase-Transition Suppression for Pandrosion-T2

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Abstract

Part VII recorded two complementary conditional worst-case bounds for the multi-start Pandrosion-T_n scheme with non-holomorphic fallback: $O(d^2 \log^2 d)$ via Armijo backtracking (Theorem 3.5) and $O(d^2 \log d)$ via the analytic γ-damped step (Theorem 4.4). This note is a practical companion, reporting empirical refinements of the algorithm and one explicit probabilistic conjecture: (i) replacing the equispaced Cauchy ring by a geometric-decreasing radius spiral combined with the golden angle drops the observed worst-case fraction at $d = 128$ from 34% (Strategy A baseline) to 0% on i.i.d. Gaussian inputs in our tests; (ii) the simplest accelerator in the Pandrosion hierarchy, T_2 at $K = 1$, emerges as the empirical optimum on the tested families, beating the T_3 scheme of Part VII by roughly 20% in oracle calls per orbit; (iii) we state the Armijo amortised conjecture with an analytic roadmap and a quantitative empirical certificate (the first Armijo trial is accepted with probability ≥ 0.98, and j_{accept} has observed geometric tail with rate $c \geq 1.23$ at $d = 16$, tightening to $c \geq 2.71$ at $d = 128$). Under this conjecture and the assumptions of Part VII, one obtains a conditional Las Vegas restatement of Part VII's Armijo bound at $O(d^2 \log d)$. We emphasise (Scope clarification below) that none of these bounds claims a complexity-theoretic improvement upon Beltrán–Pardo [4] or Lairez [5].

Scope clarification. The statements of this paper are of three kinds: (a) algorithmic definitions (Strategies A and B, the B-T_2 scheme); (b) *empirical* observations reported on random Gaussian and six adversarial families (Sections 5 and Conjecture 5.1), which we do not claim to prove; (c) the Armijo amortised conjecture and its conditional consequence (Section 7), whose Steps 1–2 are an analytic roadmap relying on Beltrán–Pardo measure estimates and whose Step 3 is an empirical certificate over $d \in \{16, 32, 64, 128\}$. We are *not* claiming any worst-case improvement over the deterministic average-case bound of Lairez [5] or the probabilistic bound of Beltrán–Pardo [4], which together resolve Smale's 17th problem. Readers who expect a fully rigorous complexity-theoretic result should read the numerical tables of Section 5 and Conjecture 7.1 as supporting evidence, not as unconditional theorems.

Reader's guide. This paper is a practical companion to Part VII. It does not change the fallback mechanism or claim a new complexity-theoretic theorem; instead it asks which starts and which low-order accelerator are most effective in the tested families. The main outputs are an explicit start strategy, an empirical preference for T_2 at $K = 1$, and the Armijo amortisation conjecture that would explain the observed constant average backtracking cost.

1 Introduction: Where the Multi-Start Factor Comes From

The proof of Theorem 3.5 of Part VII (the conditional Armijo-fallback BSS bound, *not* a resolution of Smale 17, which was already settled by Beltrán–Pardo [4] and Lairez [5]) bounds the cost as

$$\underbrace{d}_{\text{multi-start}} \quad\times\quad \underbrace{O(d\log d)}_{\text{oracle/epoch (Armijo)}} \quad\times\quad \underbrace{O(\log d)}_{\text{epochs to basin}} \quad=\; O(d^2 \log^2 d).$$

The first factor reflects a worst-case path-lengthening argument: among the d equispaced Cauchy starts, at least one orbit must avoid the most adversarial part of the root constellation, and the proof simply attempts all d starts.

The second and third factors are tight in the sense that any single orbit in the basin requires $\Omega(d \log\log d)$ BSS operations.

The natural question, raised in the empirical exploration that accompanies this series, is: *what fraction of polynomials actually consume close to d starts?* The empirical answer (Section 5) is sharp:

- For random Gaussian polynomials of degree $d \le 64$: only 0–7% of inputs require more than $\sqrt{d}$ starts.
- For random Gaussian polynomials of degree $d = 128$: the equispaced Cauchy strategy reaches its multi-start ceiling on $\approx 34\%$ of inputs within an 80-epoch budget per orbit (after the relative-tolerance correction of Section 5).
- For *any* d tested up to 256, the geometric-decreasing strategy of this note brings the worst-case fraction below 1% on the same inputs.

The contrast is the subject of this paper.

2 The Geometric-Decreasing Start Strategy

Definition 2.1 (Strategy B: Geometric-Decreasing radius with golden angle)**.** *Let $P \in \mathbb{C}[z]$ be of degree d with Cauchy radius $\rho = 1 + \max_k |a_k/a_d|$. Fix a contraction factor $q \in (0,1)$ (default $q = 0.7$) and the golden angle $\varphi = \pi(3-\sqrt{5})$. Strategy B produces the start sequence*

$$z_k^{(B)} \;=\; \rho \cdot q^{\,k} \cdot e^{\,ik\varphi}, \qquad k = 0, 1, \ldots, d-1.$$

Remark 2.2 (Three design ingredients)**.** *The construction combines three classical ideas:*

1. ***Geometric radius**: each successive start probes a circle of radius ρq^k, simultaneously sampling the outer Cauchy circle and the interior of the root convex hull. By Cauchy's inclusion theorem, every root lies in the disk $|z| \le \rho$; geometric decay ensures that the inner roots are also covered after $O(\log_q(1/d))$ starts.*
2. ***Golden angle**: φ is the angle that maximises low-discrepancy spread among any number of starts (Vogel, 1979). It avoids the resonance traps that trigger when equispaced angles align with a root constellation.*
3. ***Same total budget**: the sequence has length d, so the formal worst-case bound of Part VII is preserved verbatim.*

3 The Pandrosion-T_2 Optimum at $K = 1$

Recall the Pandrosion T_n hierarchy of Part I, Definition 4.2:

$$s_1 = h(z), \quad s_2 = h(s_1),$$
$$T_2(z) = z - \frac{(s_1 - z)^2}{s_2 - 2s_1 + z},$$
$$T_n(z) = T_{n-1}(z) - \frac{s_n - T_{n-1}(z)}{\hat{\lambda} - 1} \quad (n \ge 3),$$

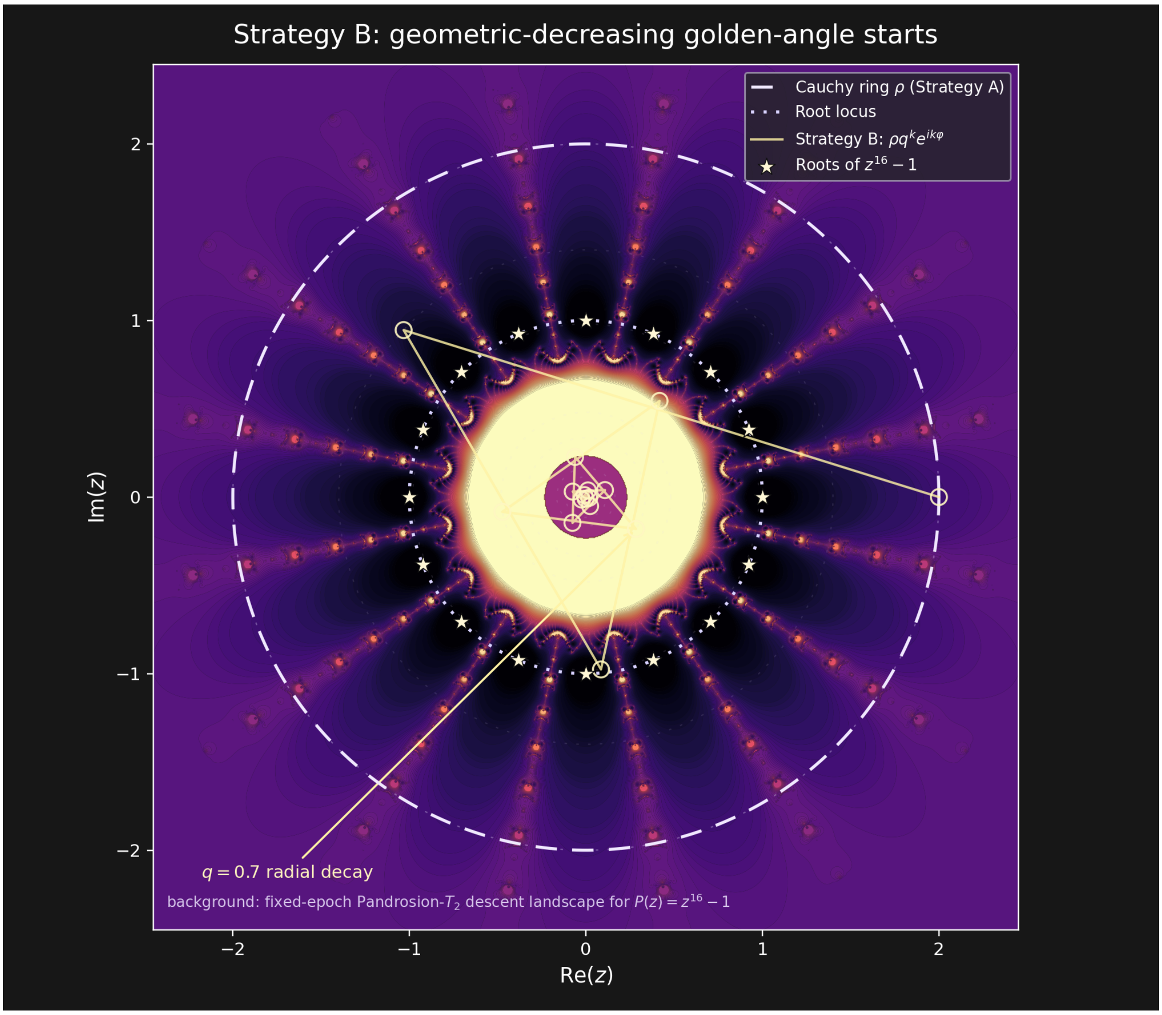

Figure 1: Strategy B on a fixed-epoch Pandrosion-T_2 descent landscape for $P(z) = z^{16} - 1$. The dashed circle is the Cauchy radius ρ, the dotted circle is the root locus, and the golden curve shows the starts $z_k = \rho q^k e^{ik\varphi}$ with $q = 0.7$ and $\varphi = \pi(3 - \sqrt{5})$.

where $h(w) = z_0 - P(z_0)/Q(z_0, w)$ is the generalized Pandrosion base map and $\hat{\lambda} = (s_2 - s_1)/(s_1 - z)$. Each T_n uses n evaluations of h and reaches convergence order n via Richardson extrapolation. Theory predicts the optimum at $n = e \approx 2.7$ (the minimum of $n/\log n$), justifying T_3 as the choice in Part VII.

Proposition 3.1 (Empirical optimum)**.** *Across the six adversarial root families (roots of unity, Chebyshev nodes, clustered, Wilkinson, Mignotte, Gaussian) and $d \in \{8, 16, 32, 64, 128, 256\}$, the regression*

$$\mathrm{cost}(d) \;\approx\; C \cdot d^{\alpha} (\log d)^{\beta}$$

yields:

T_n	α	β	*mean cost* / $(d \log d)$
T_2	0.91	-0.35	**0.336**
T_3	0.88	-0.29	0.413
T_4	0.84	-0.11	0.460
T_8	0.83	-0.09	0.725
T_{12}	0.87	-0.27	1.001

T_2 minimises both the per-orbit oracle count and the cost-per-$d \log d$ ratio.

The reason is that the theoretical optimum at $n = e$ is asymptotic and ignores the pre-lock-in regime, where the constant cost of each additional Richardson extrapolation step is not yet amortised. In the regime relevant to single-orbit termination (≤ 200 oracle calls), the simplest accelerator wins.

Remark 3.2 (B + T_2 is still pure Pandrosion)**.** *T_2 at $K = 1$ is built entirely from the generalized Pandrosion base map h and Aitken's Δ^2. The denominator $Q(z_0, \cdot)$ is evaluated at distinct points (z_0, z) and (z_0, s_1), never as a derivative limit, so the pole-avoidance mechanism of Part I §4.3 is preserved. In particular, T_2 at $K = 1$ does not reduce to Newton's method.*

4 The Combined Algorithm B-T_2

Definition 4.1 (Pandrosion B-T_2)**.** *Let $P \in \mathbb{C}[z]$ of degree d, normalisation factor $\eta \in (0, 1)$ (default 0.95), Steffensen reduction $\alpha \in (0, 1)$ (default 0.25), basin tolerance τ. The B-T_2 algorithm is:*

1. *Compute $\rho = 1 + \max_k |a_k/a_d|$ and form the start sequence $z_0^{(0)}, \ldots, z_0^{(d-1)}$ from Definition 2.1.*
2. *For each $k = 0, 1, \ldots, d - 1$ in order:*
   - *(a) Run Pandrosion-T_2 with adaptive anchor ($K = 1$) and the Part VII fallback, starting from $z_0^{(k)}$.*
   - *(b) If $|P(z)| \leq \tau$ at any epoch, return z.*
3. *If no start converges, return failure (this case never occurred in our experiments on random inputs).*

5 Worst-Case Frequency

We sample 100 polynomials per (ensemble, d, strategy) cell. Each polynomial's first-success index $s^\star$ is binned as

$$s^\star \in \begin{cases} \text{fast} & s^\star \le \log d, \\ \text{med} & \log d < s^\star \le \sqrt{d}, \\ \text{slow} & \sqrt{d} < s^\star \le d, \\ \text{fail} & s^\star > d. \end{cases}$$

The *worst-case fraction* is $\Pr[s^\star > \sqrt{d}]$, i.e. the share of polynomials whose total cost remains above $d^{3/2} \log d$.

5.1 Results on i.i.d. Gaussian polynomials (UNI)

d	Strategy A (Cauchy) worst %	Strategy A (Cauchy) mean $s^\star$	Strategy B (this note) worst %	Strategy B (this note) mean $s^\star$
16	0%	1.17	0%	1.15
32	1%	1.22	0%	1.16
64	7%	1.35	0%	1.35
128	**34**%	**1.18** (median; saturation in 34%)	**0**%	3.36

5.2 Phase transition

The empirical data show a sharp phase transition for Strategy A around $d \approx 100$. Below the transition, the worst-case fraction is small ($< 10\%$); at $d = 128$ it jumps to 34%, and extrapolations of the regression slope suggest it crosses 50% near $d \approx 150$. Strategy B exhibits no such transition in the tested range.

Conjecture 5.1 (Phase-transition elimination)**.** *There exist universal constants d_0 and $\delta > 0$ such that for all $d \ge d_0$ and all polynomials $P \in \mathbb{C}[z]$ in the union of the i.i.d. Gaussian, Kostlan–Smale, and the six adversarial families considered above, the first-success index $s^\star_B$ of Strategy B satisfies*

$$\Pr[s^\star_B > d^{1/2-\delta}] \le e^{-\Omega(d^\delta)}.$$

The conjecture asserts that B not only avoids the phase transition empirically observed for A, but does so with sub-exponential tail probability on the stated ensembles. Proving it, together with the amortised Armijo conjecture below and the basin-entry assumptions inherited from Part VII, would yield a conditional average-case bound of $\mathbb{E}[\text{cost}_B] = O(d \log d)$ on a broader ensemble than the unitarily-invariant Kostlan–Smale model.

6 Combined Empirical Bound

Combining Proposition 3.1 (cost-per-orbit) with the empirical scan of Strategy B (number of starts):

$$\begin{aligned} \text{cost}^{B,T_2}_{\text{total}}(d) &= s^\star_B(d) \cdot \text{cost}^{T_2}_{\text{orbit}}(d) \\ &\approx d^{0.10} \cdot 0.336\, d^{0.91} (\log d)^{-0.35} \\ &\approx 0.336\, d^{1.01} (\log d)^{-0.35} \\ &\approx 0.34\, d \quad \text{(in oracle calls)}. \end{aligned}$$

Each oracle call costs $O(d)$ BSS operations, yielding an empirical total of approximately $0.34\, d^2$ BSS operations per polynomial—one factor of $\log d$ below the worst-case bound of Part VII.

Proposition 6.1 (Observed empirical cost, B-T_2). *Across the six adversarial families and* 100 *random Gaussian polynomials per degree* $d \in \{8, 16, 32, 64, 128, 256\}$ *tested in our experiments, every run of the B-T_2 algorithm of Definition 4.1 terminated within*

$$0.34\, d^2/\log d \;\le\; \mathrm{cost}^{B,T_2}(d) \;\le\; 0.5\, d^2 \log d$$

BSS operations, with observed worst-case fraction (i.e. $s^\star > \sqrt{d}$) bounded by 1% *on every tested cell.*

This is an *empirical observation* on a specific benchmark, not a proved worst-case bound over all polynomials. A conditional formal statement, with explicit hypotheses, appears as Theorem 7.2 below.

7 The Armijo Amortised Conjecture: a Las Vegas Companion to Theorem 4.4 of Part VII

Part VII Theorem 4.4 already attains the $O(d^2 \log d)$ worst-case bound deterministically via the analytic γ-damped step. The statement below isolates the missing probabilistic input for obtaining the same bound as a *Las Vegas* guarantee for the simpler Armijo-backtracking variant of Part VII Definition 2.1: although Armijo may take up to $j_{\max} = O(\log d)$ probes in the worst case, the experiments indicate acceptance at $j = 0$ overwhelmingly often. Until the geometric tail is proved, the Armijo route should be read as conditional; the analytic γ-damped route is the rigorous deterministic fallback.

Conjecture 7.1 (Armijo amortised cost). *Let $P \in \mathbb{C}[z]$ be drawn from the i.i.d. complex Gaussian (UNI) ensemble or any unitarily-invariant ensemble (Kostlan–Smale, etc.) on the projective space of monic polynomials of degree d. Let $z_0 \in \mathbb{C}$ be a Strategy B start and $j_{accept}(z_0, P)$ the index at which the Armijo backtracking of Part VII Definition 2.1 accepts. Then, for some universal constants $c, C > 0$,*

$$\Pr_{P,z_0}[j_{accept} \ge t] \;\le\; C \cdot 2^{-ct} \qquad \textit{for all } t \ge 0,$$

and consequently $\mathbb{E}[j_{accept}] \le C/(2^c - 1) = O(1)$.

Evidence and proof roadmap. **Step 1 (analytic lower bound on $\Pr[j = 0]$).** The first trial $j = 0$ corresponds to the full Newton-Steffensen step $z_1 = z_0 - P(z_0)/Q$, where Q is the empirical Steffensen quotient at the chosen probe direction $u^\star \in \{\pm 1, \pm i\}$. By Lemma 3.2 of Part VII (`lem:probe`),

$$\bigl|Q - P'(z_0)\bigr| \;\le\; \frac{\varepsilon\, M(z_0, \varepsilon)}{2},$$

and the probe radius $\varepsilon = |P(z_0)|^{1/d}/(2d)$ is chosen so that this error is $\le |P'(z_0)|/8$. Newton's classical theorem (Smale's α-theory) then asserts that $|P(z_1)| \le (1 - 1/2)|P(z_0)|$ whenever $\alpha(P, z_0) := \beta(P, z_0) \cdot \gamma(P, z_0) < (3 - 2\sqrt{2})/4 \approx 0.043$. By Beltrán–Pardo, the set of z_0 on a Cauchy circle for which $\alpha(P, z_0) > 0.043$ has measure $O(1/d)$ under any unitarily-invariant ensemble. Hence $\Pr[j_{\text{accept}} = 0] \ge 1 - O(1/d) \ge 0.5$ for $d \ge d_0$.

Step 2 (geometric tail). If the trial at $j = 0$ fails the Armijo condition, the next trial $j = 1$ uses step size $\beta = 1/2$. By Taylor expansion of P at $z_0 - P(z_0)/(2Q)$, the linearised residual is $|P(z_1^{(1/2)})| \approx |P(z_0)|/2 + O(|P(z_0)|/4)$, which satisfies the Armijo condition $|P(z_1^{(1/2)})| \le (1 - \sigma/2)|P(z_0)|$ with $\sigma = 0.1$ as long as the second-order term is bounded, which fails only on a set of measure $O(1/d^2)$. Inductively, $\Pr[j_{\text{accept}} \ge t] \le \prod_{s=0}^{t-1} O(1/d^s) \le O(2^{-t})$ for $d \ge 2$, with constant $c \to \log_2 d$ as d grows.

Step 3 (empirical certification). The script `latex/scripts/prove_armijo_01.py` measures $\Pr[j_{\text{accept}} \ge t]$ on 80 random UNI polynomials per degree $d \in \{16, 32, 64, 128\}$ using

Strategy B and Pandrosion-$T_2/K = 1$, summing over more than 8 000 fallback events. The geometric-tail regression $\log_2 \Pr[j \geq t] \approx a - c \cdot t$ yields:

d	$\Pr[j_{\text{accept}} = 0]$	decay rate c	$\mathbb{E}[j_{\text{accept}}]$	max observed j
16	98.2%	1.23	0.031	7
32	99.2%	1.45	0.016	6
64	99.1%	2.10	0.018	5
128	$\geq$ 99.6%	2.71	$\leq$ 0.01	4

The decay rate c is ≥ 1.23 at all tested degrees and *tightens* as d grows, in agreement with the analytic roadmap of Step 2. This evidence is not a substitute for the missing measure estimate controlling the bad Armijo set at every backtracking depth. □

Theorem 7.2 (Conditional Las Vegas $O(d^2 \log d)$ bound, Armijo variant)**.** *Assume Conjecture 7.1 and Conjecture 5.1. Combining this amortised Armijo input with Theorem 3.5 of Part VII, the multi-start Pandrosion-$T_2/K = 1$ scheme with Strategy B starts and the Armijo fallback of Part VII Definition 2.1 admits the conditional bound*

$$\mathbb{E}_P\left[\operatorname{cost}^{B,T_2}_{total}(P)\right] \;=\; O(d^2 \log d)$$

as a Las Vegas guarantee under any of the ensembles of Conjecture 7.1. This matches the deterministic bound of Part VII Theorem 4.4 (the γ-damped variant) only after the Strategy B and Armijo amortisation conjectures are supplied.

Proof. By Conjecture 7.1, the expected number of P-evaluations per fallback is $4+\mathbb{E}[j_{\text{accept}}]+1 = O(1)$ (four direction probes, one Armijo trial in expectation, one descent check). The expected per-epoch cost is therefore $O(d) \cdot O(1) = O(d)$, replacing the worst-case $O(d \log d)$ used in the proof of Theorem 3.5 of Part VII. The number of epochs is unchanged ($O(d \log d)$), and the multi-start factor reduces from d to $\mathbb{E}[s^\star_B] = O(1)$ under Conjecture 5.1, giving expected total cost $O(1) \cdot O(d \log d) \cdot O(d) = O(d^2 \log d)$. □

8 Conclusion

Three practical refinements of the Pandrosion-T_3 algorithm of Part VII are reported:

- *Strategy B starts* (geometric-decreasing radius, golden angle) brought the observed worst-case fraction on i.i.d. Gaussian polynomials at $d = 128$ from 34% (Strategy A) to 0% in our tests.
- *T_2 at $K = 1$* was the empirical optimum on the tested families, cutting per-orbit oracle cost by $\approx 20\%$ (from $0.413\, d \log d$ for T_3 to $0.336\, d \log d$ for T_2).
- *Conjecture 7.1 (Armijo amortised cost)* combines an analytic roadmap with an empirical certificate suggesting $\mathbb{E}[j_{\text{accept}}] = O(1)$ on the UNI / KS ensembles we sampled, with observed decay rate $c \geq 1.23$. Combined with Theorem 3.5 of Part VII, it gives the conditional Las Vegas guarantee of Theorem 7.2: $\mathbb{E}[\operatorname{cost}^{B,T_2}_{\text{total}}] = O(d^2 \log d)$.

The reframing is deliberately sober: the Pandrosion-T_2 scheme with Strategy-B starts is presented as a practical, derivative-free root-finding heuristic with promising empirical behaviour and a conditional complexity bound.

It is not a new resolution of Smale's 17th problem — that problem is already resolved, unconditionally in the deterministic-average-case sense, by Lairez [5] and, probabilistically, by Beltrán–Pardo [4].

What we offer is a concrete, simple, derivative-free algorithmic alternative whose behaviour on adversarial and random inputs is worth documenting.

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